2023-Winter Paper 3 – Subspecialities of Urology

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Q1. Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis.

Type: LAQ (25 marks) (medical management also answered in 2019-Summer Paper 2, Q3e)

Introduction

- RPF (retroperitoneal fibrosis) is a rare inflammatory fibrosis encasing the aorta, IVC (inferior vena cava) and ureters.
- Incidence about 1 per 200,000–500,000 a year; age 40–60; male:female 2–3:1.

Etiology

Idiopathic (about 70%)	Secondary (about 30%)
Part of **chronic periaortitis**, a large-vessel vasculitis	**Drugs** (commonest identifiable cause): methysergide and other ergots, β-blockers, phenacetin
Proposed antigen: **ceroid** (oxidised lipid in plaque)	**Malignancy:** lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma
Probably autoimmune: membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis; ANA (antinuclear antibody) positivity predicts relapse	Radiotherapy; asbestos
Part of multifocal fibrosclerosis: sclerosing mediastinitis and cholangitis, orbital pseudotumour, Riedel thyroiditis	Infection: tuberculosis, actinomyces, gonorrhoea, schistosomiasis

- Many idiopathic cases are now classed as **IgG4 (immunoglobulin G4)-related disease**.

Pathology

- **Gross:** flat, tan, dense plaque centred on the **distal aorta at L4–L5**, wrapping the ureters; may invade ureter or psoas.
- **Histology changes with stage:**
  - Early (active): collagen with capillary proliferation and lymphocytes, plasma cells and fibroblasts.
  - Late (burnt-out): acellular, avascular sheets of collagen.
- **Trap:** malignant RPF looks identical except for small islands of tumour cells, so take **deep, multiple biopsies**.

- **Mechanism of obstruction:** extrinsic compression and loss of ureteric peristalsis; the lumen stays patent.

Clinical features

- **Dull lower back or flank pain** (commonest); weight loss, anorexia, nausea, malaise, fever, hypertension.
- About half present with **uraemia** or oliguria/anuria; symptoms last 4–6 months before diagnosis.
- **Extension:** IVC (leg oedema, DVT [deep venous thrombosis]); renal vein (gross haematuria); spermatic cord and scrotum. Examination is usually unrevealing.

Diagnostic evaluation

Laboratory

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein) in half to two-thirds; leucocytosis; anaemia; raised creatinine; ANA; serum IgG4.

Imaging

Test	Findings and role
CT (computed tomography) / CT urography	Homogeneous soft-tissue mass enveloping aorta, IVC and ureters; **medial deviation** of proximal and mid-ureters with smooth tapering
Retrograde pyelography	When renal function is poor
MRI (magnetic resonance imaging)	Low T1; **high T2 signal means active disease**; enhancement falls with treatment
PET-CT (positron emission tomography)	**Most sensitive test of activity**; guides biopsy and follow-up
Renography	Split function (repair or nephrectomy)

- **Biopsy** (CT-guided or at ureterolysis) if malignancy is suspected; may be omitted with no malignancy history, no lymphadenopathy and typical imaging.

Treatment

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RPF with hydronephrosis
  -> Relieve obstruction (stents; nephrostomy if anuric or septic)
  -> Treat post-obstructive diuresis
  -> Stop culprit drug; exclude malignancy
  -> Active disease: steroids (+/- second-line drugs)
  -> Failure or burnt-out fibrosis: ureterolysis
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1. Initial management

- Emergent **stents or PCN (percutaneous nephrostomy)**; nephrostomy if critically ill with little urine. Replace fluid during **post-obstructive diuresis**.

2. Medical therapy

Agent	Role and regimen
Prednisone (first line)	60 mg daily (about 1 mg/kg), tapered to 5 mg, up to 2 years. About 80% respond; relapse on tapering 25–50%
Azathioprine, mycophenolate mofetil, cyclophosphamide, ciclosporin, colchicine	Second line or steroid-sparing
Tamoxifen; medroxyprogesterone	Thought to inhibit fibroblasts
Rituximab	Refractory or IgG4-related disease

- Steroids work on the cellular, active phase: pain settles in days and the mass shrinks.

- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen.
- Drainage plus medical therapy clears obstruction in about two-thirds; only about **15% need ureterolysis**.

3. Surgical ureterolysis

- For failed medical therapy, burnt-out fibrosis or tissue diagnosis. **Bilateral**, even if hydronephrosis is unilateral; open, laparoscopic or robotic.
- Steps:
  1. Pre-place stents to identify the ureters.
  2. Mobilise the colon; take deep biopsies for frozen section.
  3. Start at the distal, non-dilated ureter.
  4. "Split and roll": right-angle clamp between ureter and plaque; incise the fibrosis over it.
  5. Close any ureterotomy with absorbable suture.
  6. **Separate the ureter from the fibrotic bed so it cannot be re-encased.**

Protection method	How
Intraperitonealisation	Ureter lateralised; peritoneum closed behind it
Lateral retroperitoneal placement	Ureter displaced laterally (equivalent results)
Omental wrap	Omentum split on the gastroepiploic arteries, wrapped round each ureter; adds blood supply; best for extensive disease

- Postoperative steroids may reduce recurrence.
- **Outcomes:** open about 94% stent-free at 1 year; laparoscopic about 90%; robotic with omental wrap about 86%.

4. Other options: long-term metallic stents (unfit); autotransplantation if ureterolysis is impossible; nephrectomy for an unrecoverable kidney.

Follow-up

- Symptoms, ESR/CRP, creatinine, ultrasound; MRI or PET-CT for activity; remove stents once ureters drain; watch steroid side effects.

Diagram to draw: periaortic plaque at L4–L5 drawing both ureters medially, with an omental wrap.

> **Key points to remember:**
> - Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
> - Medially deviated ureters; PET-CT best shows activity; biopsy deep.
> - Drain first, then prednisone (about 80% respond; Vaglio 2011 beat tamoxifen).
> - About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

Q2. Discuss Etiology, Diagnosis and Management of Upper ureteric injury.

Type: LAQ (25 marks)

Assumed meaning: injury to the upper (proximal) ureter, from the PUJ (pelviureteric junction) to the pelvic brim, from external trauma or surgery.

Introduction

- Ureteric trauma is uncommon: 1–2.5% of genitourinary trauma. **Iatrogenic injury is commoner overall** than external trauma.
- **The proximal ureter is the commonest site of external injury.** The distal ureter is the commonest iatrogenic site in non-urological surgery.
- Injury is often **missed** and declares itself later as urinoma, sepsis, stricture or fistula.

Etiology

1. External trauma

- **Penetrating** (most): gunshot wounds up to about 90%. Blast damages the intramural blood supply up to 2 cm from the wound.
- **Blunt** (up to about 40%): rapid deceleration avulses the ureter at the **PUJ**; linked to lumbar transverse process fractures.
- **Over 90% have associated injuries** (small bowel commonest, colon, vessels, kidney 10–28%). Mortality 6–9% reflects these.

2. Iatrogenic

Source	Upper-ureter relevance
Ureteroscopy	Commonest iatrogenic cause in some series: abrasion, perforation (0.2–2%), avulsion (<0.3%); stricture 0.3–3%. Risk: proximal stone, impacted stone, long operation, large sheath, inexperience
Partial nephrectomy, thermal ablation	Proximal stricture months later (lower-pole and hilar tumours)
Vascular (aortic) surgery	Devascularisation; up to 85% missed initially
Colorectal, spinal, retroperitoneal surgery	Ligation, transection, cautery

- **Mechanisms:** ligation, transection, avulsion, devascularisation, thermal injury.

Classification: AAST (American Association for the Surgery of Trauma) ureter scale

Grade	Description
I	Contusion or haematoma without devascularisation
II	Laceration <50% of circumference
III	Laceration ≥50%
IV	Complete transection with <2 cm devascularisation
V	Avulsion with >2 cm devascularisation

- Advance one grade for bilateral injury, up to grade III.

Diagnosis

Suspicion

- Penetrating flank or abdominal wound; high-energy deceleration.
- **Haematuria is absent in 25–45%**, so its absence does not exclude injury.
- Late or postoperative signs: fever, ileus, flank pain, leucocytosis, rising creatinine, drain or wound urine leak, urinoma.

Intraoperative

- Inspect the ureter at laparotomy; IV (intravenous) methylene blue or indigo carmine; on-table retrograde pyelogram. About 11% are missed at trauma laparotomy.

Imaging

Test	Role
CT urography with 10-minute delayed images	Best test in stable patients. Signs: extravasation, non-opacified distal ureter, delayed nephrogram, urinoma
Retrograde pyelography	Most sensitive; shows site; allows stenting at once
Antegrade nephrostogram	When retrograde fails or gap is wide
One-shot IVU (intravenous urography)	Non-diagnostic in 33–100%; never rely on it

- **Drain-fluid creatinine** far above serum creatinine confirms urine.

Management

Principles

- **Stability decides timing; level and length decide the operation.**
- Repair: mobilise sparing adventitia; debride to bleeding edges; **spatulated, tension-free, watertight, stented** anastomosis with fine absorbable suture; drain; retroperitonealise; omental wrap for blast or vascular injury.
- **No bowel interposition or autotransplantation acutely.**

```
Ureteric injury suspected
 |-- Unstable -> damage control: ligate + later PCN,
 |               or exteriorised stent / cutaneous ureterostomy
 |-- Stable, found at surgery -> immediate repair over stent
 |-- Found late / postoperatively
       -> Attempt retrograde stent (+/- PCN, drain)
       -> Fails -> PCN; delayed definitive repair
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Repair by grade

- **Grade I–II (contusion, partial):** double-J stent for about 6 weeks; or close a laceration transversely over a stent. Excise non-viable contused ureter.
- **Grade III–V:** formal reconstruction.

Options for the upper ureter

Procedure	Key points
Ureteroureterostomy	Commonest acute repair; short defects (2–3 cm); spatulate 5–6 mm, 180° apart; 5-0 absorbable over double-J; up to 90% success; leak 10–24%, stricture up to 12%
Ureteropyelostomy (reimplant into pelvis)	PUJ avulsion; spatulated
Ureterocalicostomy	Pelvis or PUJ destroyed; lower-pole parenchyma resected
Renal descensus	Mobilise kidney downwards; gains up to 8 cm

Procedure	Key points
TUU (transureteroureterostomy)	Delayed; long loss when the bladder cannot reach; 90–97% success. Avoid with stones, RPF (retroperitoneal fibrosis), TB (tuberculosis), urothelial cancer or radiation
Ileal ureter / Yang–Monti	Long or panureteric loss; 81–100% success. Avoid if creatinine >2 mg/dL, bladder dysfunction, inflammatory bowel disease, radiation enteritis. Acidosis about 20%
Appendiceal interposition	Right side; about 92% short-term success
Buccal mucosa graft ureteroplasty	Delayed long strictures; onlay with omental wrap, often robotic; about 87% success
Autotransplantation	Last resort before nephrectomy; 5–10% renal units lost
Nephrectomy	Rarely; unstable with unreconstructable kidney, or poor function

Iatrogenic endoscopic injury

- Stop the procedure and place a **stent** (± nephrostomy); later reconstruction if stricture develops.

Complications and follow-up

- Urinoma, abscess, leak, fistula, stricture, loss of renal function.
- Persistent leak: check stent position, take drain off suction, catheterise the bladder, then nephrostomy.
- **Follow-up:** ultrasound or diuretic renogram at 4–12 weeks and at 1 year; late stenosis can be silent.

Diagram to draw: spatulated ureteroureterostomy over a double-J stent.

> **Key points to remember:**
> - Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
> - AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
> - CT with 10-minute delays; retrograde pyelogram most sensitive.
> - Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Q3a. Premature ejaculation.

Type: SAQ (10 marks)

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed:
  1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired).
  2. Inability to delay ejaculation on all or nearly all penetrations.
  3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

Aetiology

- **Lifelong:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.
- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Treat the cause in acquired PE (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug)	30–60 mg, 1–3 h before intercourse
Paroxetine (strongest delay), sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine (spray licensed in Europe 2014)	5–30 min before; condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI
Tramadol	Low evidence; addiction risk

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.

Diagram to draw: office algorithm (confirm PE, then treat cause or classify lifelong/acquired).

Key points to remember:

- ISSM: about 1 min (lifelong), about 3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Q3b. Desmopressin.

Type: SAQ (10 marks)

Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).
- Deamination and D-arginine substitution prolong action and make it a **selective V2 agonist without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells → cAMP (cyclic adenosine monophosphate) → **aquaporin-2** insertion → water reabsorption → less, concentrated urine.
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

Formulations

Form	Dose
Tablet	0.2–0.4 mg at bedtime (enuresis)
Oral lyophilisate (melt)	120–240 µg, about 1 h before bed
Low-dose nasal spray (nocturia)	0.83 or 1.66 µg, 30 min before bed
Sublingual ODT (orally disintegrating tablet)	27.7 µg women, 55.3 µg men
IV / subcutaneous	0.3 µg/kg (bleeding)

Uses in urology

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. Rule of thirds: one-third full, one-third partial, one-third no response. Acts at once but relapses on stopping. Restrict fluids in the evening.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below 65; low dose above 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired in Wilms tumour). Tachyphylaxis after about 3 doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.
- Most events occur in the **first 2–3 weeks**.

Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

Diagram to draw: V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

Key points to remember:

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

Q3c. Snodgrass repair for Hypospadias.

Type: SAQ (10 marks)

Introduction

- **TIP (tubularised incised plate) urethroplasty**, described by Snodgrass in 1994 (based on Rich's hinging incision, 1989).
- Used for **over 90% of distal and about 80% of midshaft repairs**.

Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate, so it can be tubularised without tension.
- The incised raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

Indications and prerequisites

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (<30°) and corrected by degloving or plication.
- Adequate plate and glans. **Glans width <14 mm** raises complications; give preoperative testosterone.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft.
- Not for severe curvature needing plate division; proximal TIP is abandoned.

Operative steps

1. Glans stay suture; tourniquet.
2. U-incision around plate and meatus; deglove; artificial erection.
3. Separate glans wings deeply from the plate.
4. **Midline incision of the plate** from the meatus to the plate end, down to corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, stopping 2–3 mm short to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier.
7. Two-layer glansplasty; mucosal collar; skin closure (Byars flaps) or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for spasm.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, meatal stenosis 3.6%, stricture 1.3%.
- **Meatal stenosis** is commoner than after Mathieu. Glans dehiscence occurs.
- Proximal and redo cases do worse; a dartos layer improves results.

Modifications

- **Dorsal inlay graft** (inner preputial skin) for a shallow plate; lateral incisions to 12 mm width.

Diagram to draw: cross-sections showing plate incision, tubularisation and dartos cover.

Key points to remember:

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Q3d. Urological Complications post Kidney Transplantation.

Type: SAQ (10 marks)

Introduction

- Urological complications are the **commonest surgical complications** (8–9.5%). Most arise from the **ureter and ureterovesical anastomosis**.
- The distal donor ureter depends on a single renal-artery blood supply, so ischaemia is the root cause. **Denervation** masks pain.
- A **routine stent** reduces leak and stricture; remove at **14–21 days** (earlier increases leak; later increases UTI [urinary tract infection]).

Complications

Complication	Key facts	Management
Urine leak (1.2–8.9%; commonest early)	Distal ureteric necrosis; lost lower-pole artery; technical	Fluid creatinine far above serum. Foley + PCN (percutaneous nephrostomy) + antegrade stent; early reimplant or native ureteroureterostomy if large
Ureteric stricture (0.5–9%; commonest beyond 3 months)	Ischaemia; long cold ischaemia, rejection, BK (BK polyomavirus) and CMV (cytomegalovirus)	Nephrostomy first; balloon or laser endoureterotomy; **open reimplant, native ureteroureterostomy or Boari flap best long-term**
Lymphocele (symptomatic 0.03–26%; peak 6 weeks)	Divided lymphatics; mTOR (mammalian target of rapamycin) inhibitors	Treat only if symptomatic; **laparoscopic peritoneal fenestration** best
VUR (vesicoureteric reflux)	50–86% radiological; <1% significant	Treat cause; endoscopic injection low grade; reimplant
Haematuria	Anastomotic; persistent after biopsy = AV fistula	Self-limiting; selective embolisation
Stones (about 1%)	Hyperparathyroidism; calcineurin inhibitors	As native kidney; watch silent obstruction
UTI (commonest infection)	Stent, reflux, steroids	Treat 14–21 days if pyelonephritis

- **Obstruction:** early is technical, oedema or compression (haematoma, lymphocele); late is stricture, stone or lymphocele.
- **Encrusted stent:** CT, nephrostomy, then PCNL (percutaneous nephrolithotomy), RIRS (retrograde intrarenal surgery) or ESWL (shock-wave lithotripsy).
- **Lower tract:** bladder capacity recovers; BPH (benign prostatic hyperplasia) treated by TURP (transurethral resection of prostate) after 3 weeks, never with a stent in place.
- **Also know:** ED (erectile dysfunction) persists in 20–50%; bladder cancer risk about threefold; RCC (renal cell carcinoma) mostly in native kidneys; renal artery stenosis and thrombosis.

Diagram to draw: transplant kidney in iliac fossa showing leak, stricture and lymphocele sites.

> **Key points to remember:**
> - Ureter is the weak point; ischaemia is the cause.
> - Leak early, stricture late; open repair best.
> - Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

Q3e. Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

Type: SAQ (10 marks)

Definition

- **CIS (carcinoma in situ, Tis):** flat, non-invasive, **high-grade** urothelial carcinoma. It is not "premalignant".
- Histology: severe atypia, loss of polarity, non-cohesive pleomorphic cells, **loss of umbrella cells** (dysplasia keeps them).
- **Types:** primary (isolated, about 10%), secondary or concurrent (with papillary tumour, about 90%).

Pathogenesis

- **Flat pathway:** dysplasia → CIS → high-grade invasive cancer.
- Driven by **TP53, RB and PTEN** loss; **no FGFR3** mutation (unlike low-grade papillary tumours).
- **Field change:** multifocal; spreads to prostatic urethra, distal ureters and upper tract.

Clinical importance

- Untreated, **40–83% progress** to muscle invasion. Primary CIS progresses in >50%.
- "CIS alone" hides invasion at cystectomy in up to 20%.
- **Second strongest prognostic factor after grade**; any CIS makes NMIBC (non-muscle-invasive bladder cancer) **high risk**.
- About 80% have **irritative symptoms**; it mimics cystitis.

Diagnosis

- **Urine cytology:** most useful in CIS, because the cells are poorly cohesive. Positive cytology with normal cystoscopy needs mapping.
- Cystoscopy: red velvety patch or normal-looking mucosa; **blue-light cystoscopy** (hexaminolevulinate) detects about one-third more CIS; NBI (narrow-band imaging).
- **Biopsies:** cold-cup from suspicious and random sites; prostatic urethra; upper tract imaging.

Management

- **BCG (bacillus Calmette–Guérin) is first line:**
  - Induction: 6 weekly instillations.
  - **Maintenance** (SWOG [Southwest Oncology Group] 2000: 3 weekly doses at 3 and 6 months, then 6-monthly for 3 years): median recurrence-free survival 76.8 vs 35.7 months.
  - Complete response up to 84%; durable in about half.
- **Persistent CIS at 3 months:** second BCG course; 30–50% more respond.
- **BCG-unresponsive** (persistent or recurrent CIS within 12 months of adequate BCG): **radical cystectomy**.
- Unfit or unwilling: **pembrolizumab** (KEYNOTE-057: complete response 41%), nadofaragene firadenovec, gemcitabine–docetaxel.
- Prostatic urethral CIS: TURP plus BCG; ductal involvement needs cystoprostatectomy.

Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly; upper-tract imaging.

Diagram to draw: flat versus papillary pathways of bladder cancer.

Key points to remember:

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive.

Q3f. Laboratory Diagnosis of Genitourinary Tuberculosis.

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 3, Q2, as part of renal TB)

Principle

- GUTB (genitourinary tuberculosis) is paucibacillary with intermittent shedding, so **multiple samples** are needed.
- **Hierarchy of proof:** positive culture or DNA → caseating granulomas on histology → clinical picture with response to treatment.

Specimen

- **Early-morning first-void urine, 3–5 samples on consecutive days**, sent at once (acidic urine kills bacilli). Avoid prior fluoroquinolones.
- Also tissue, pus, FNAC (fine-needle aspiration cytology) or semen.

Tests

Test	Method	Comment
Urine routine	Microscopy	**Sterile pyuria** ± haematuria in >90%
ZN (Ziehl–Neelsen) smear	Carbol fuchsin, acid-alcohol, methylene blue: red bacilli. Auramine fluorescence is more sensitive	Sensitivity <50%; saprophytic mycobacteria give false positives
Culture: gold standard	Decontaminate and concentrate. **LJ (Löwenstein–Jensen)** solid: 4–6 weeks. **MGIT (mycobacteria growth indicator tube)** liquid: about 10 days	80–90% sensitivity with ideal technique; gives DST (drug-susceptibility testing); monitors treatment
NAAT (nucleic acid amplification test): GeneXpert MTB/RIF	Automated cartridge PCR (polymerase chain reaction); about 2 hours	Detects MTB and **rifampicin resistance** (MDR [multidrug-resistant] TB surrogate). **Xpert Ultra** is more sensitive. Cannot monitor treatment
TrueNat	Indian chip-based NAAT	NTEP (National TB Elimination Programme)
LPA (line probe assay)	DNA strip hybridisation	Rifampicin, isoniazid or second-line drug resistance
LF-LAM (lateral-flow lipoarabinomannan)	Urine antigen strip, <30 min	HIV (human immunodeficiency virus) with CD4 <200 or advanced disease (WHO 2019)
Histopathology	Biopsy (bladder, epididymis)	Caseating granulomas; NAAT on tissue raises yield
TST (tuberculin skin test)	5 TU (tuberculin units) PPD intradermal; read induration at 48–72 h	Shows infection, not disease; BCG false positive
IGRA (interferon-γ release assay)	ESAT-6/CFP-10 blood test (QuantiFERON, T-SPOT)	Not affected by BCG; cannot separate latent and active

Supportive tests

- ESR, blood count, renal function, HIV, blood sugar; sputum and chest radiograph (lung TB in up to one-fifth).

Diagram to draw: stepwise diagnostic flowchart from sterile pyuria to culture and DST.

> **Key points to remember:**
> - 3–5 early-morning urine cultures; culture is the gold standard.
> - GeneXpert detects rifampicin resistance in 2 hours; cannot monitor treatment.
> - TST and IGRA show infection, not active disease.

Rapid Revision Sheet – 2023-Winter Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis. (LAQ)

- Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
- Medially deviated ureters; PET-CT best shows activity; biopsy deep.
- Drain first, then prednisone (about 80% respond; Vaglio 2011 beat tamoxifen).
- About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

Q2. Discuss Etiology, Diagnosis and Management of Upper ureteric injury. (LAQ)

- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram most sensitive.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Q3a. Premature ejaculation. (SAQ)

- ISSM: about 1 min (lifelong), about 3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Q3b. Desmopressin. (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early.

Q3c. Snodgrass repair for Hypospadias. (SAQ)

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Q3d. Urological Complications post Kidney Transplantation. (SAQ)

- Ureter is the weak point; ischaemia is the cause.
- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

Q3e. Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder. (SAQ)

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive.

Q3f. Laboratory Diagnosis of Genitourinary Tuberculosis. (SAQ)

- 3–5 early-morning urine cultures; culture is the gold standard.
- GeneXpert detects rifampicin resistance in 2 hours; cannot monitor treatment.
- TST and IGRA show infection, not active disease.